

Course Code: **IDC 222**



Scientific Computing & Data Visualization

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EXERCISES

Problem 1: Kinetic Energy of a Body

Write a C++ program to calculate the kinetic energy of a moving body using the

Formula : $KE = \frac{1}{2}mv^2$

Where:

- m** - mass of the body (in kg)
- v** - velocity of the body (in m/s)

OUTPUT FORMAT:

Enter mass of the body (kg): **4**

Enter velocity of the body (m/s): **5**

Kinetic Energy = **50 Joules**

Program:

```
#include <iostream>
using namespace std;

int main()
{
    float mass, velocity, kineticEnergy;

    cout << "Enter mass of the body (kg): ";
    cin >> mass;

    cout << "Enter velocity of the body (m/s): ";
    cin >> velocity;

    kineticEnergy = 0.5 * mass * velocity * velocity;

    cout << "Kinetic Energy = " << kineticEnergy << " Joules";

    return 0;
}
```

Problem 2: Time Period of a Simple Pendulum

Write a C++ program to calculate the time period of a simple pendulum using the

Formula :
$$T = 2\pi \sqrt{\frac{l}{g}}$$

Take $\pi = 3.14$ [pi]

Where:

T = time period (in seconds)

l = length of the pendulum (in meters)

g = acceleration due to gravity $= 9.8 \text{ m/s}^2$

OUTPUT FORMAT:

Enter length of the pendulum (m): 1.5

Time Period = 2.45693 seconds

Program:

```
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    float length, timePeriod;
    float pi = 3.14, g = 9.8;

    cout << "Enter length of the pendulum (m): ";
    cin >> length;

    timePeriod = 2 * 3.14 * sqrt(length / g);

    cout << "Time Period = " << timePeriod << " seconds";

    return 0;
}
```

Problem 3: Range of a Projectile

Write a C++ program to calculate the range of a projectile using the

Formula :
$$R = \frac{v^2 \sin(2\theta)}{g}$$

Where:

R = range of the projectile (in meters)

v = initial velocity (in m/s)

θ = angle of projection (in degrees)

g = acceleration due to gravity $= 9.8 \text{ m/s}^2$

OUTPUT FORMAT:

Enter initial velocity (m/s): 20

Enter angle of projection (degrees): 45

Range of the projectile = 40.8163 meters

Note : Angles in `sin()` must be in radians, not degrees: $\text{radians} = \text{degrees} \times \frac{\pi}{180}$

Program:

```
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    float velocity, angle, range;
    const float g = 9.8;

    cout << "Enter initial velocity (m/s): ";
    cin >> velocity;
    cout << "Enter angle of projection (degrees): ";
    cin >> angle;

    float angleRad = angle * 3.14 / 180.0; // angle to radians conversion

    range = (velocity * velocity * sin(2 * angleRad)) / g;

    cout << "Range of the projectile = " << range << " meters";
    return 0;
}
```


Control Statements



5. Control Statements

- selective or repetitive execution of certain part of the program is achieved with the help of certain part of the statements known as *control transfer statements* or *control structures*.

5.1 Selection Statements

- Selection statements provide decision making capability to a program.
- They alter the normal statements provide decision making capability to a program.
- They alter the normal flow of control based on a condition and hence they are also known as **conditional statements** or **decision statements**.
- There are **two selection statements** in C++: **if** and **switch**.
- The **if statement** has many versions such as simple **if**, **if-else**, **if-else-if**.
- Apart from these statements, conditional expression, i.e., expressions using conditional operator may be used for implementing the selection control structure.



C++ has the following conditional statements:

- Use **if** to specify a block of code to be executed, if a specified condition is true
- Use **else** to specify a block of code to be executed, if the same condition is false
- Use **else if** to specify a new condition to test, if the first condition is false
- Use **switch** to specify many alternative blocks of code to be executed



The if statement

Use the `if` statement to specify a block of C++ code to be executed if a condition is `true`.

Syntax:

```
if (condition)
{
    // block of code to be executed if the condition is true
}
```

Note that `if` is in lowercase letters. Uppercase letters (If or IF) will generate an error.



In the example below, we test two values to find out if 20 is greater than 18.

If the condition is **true**, print some text:

Example

```
if (20 > 18)
{
    cout << "20 is greater than 18";
}
```

Example

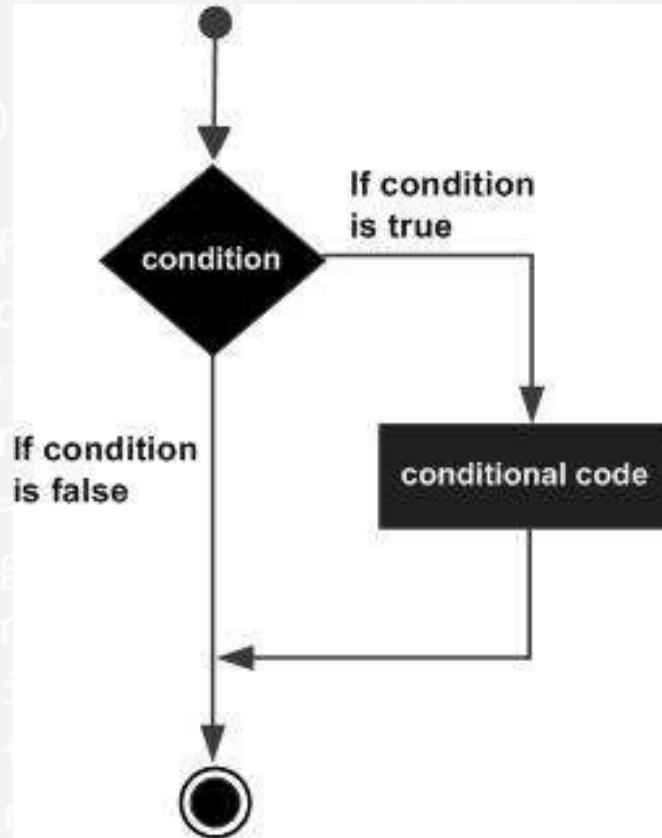
```
int x = 20;
int y = 18;
if (x > y)
{
    cout << "x is greater than y";
}
```

Exercise:

Print "Hello World" if **x** is greater than **y**.



flowchart of if statement

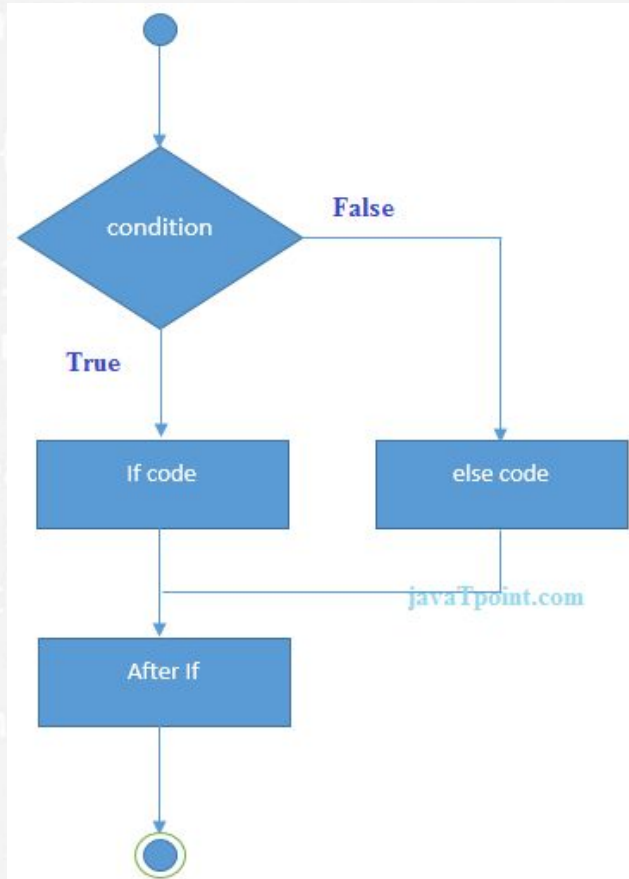


Contd...

- The statement may be a simple statement or a compound statement or a null statement.
- A sequence of C++ statements enclosed on a pair of braces { } is called a **compound statement**.
- In a compound statement, each individual statement ends in a semi-colon.
- The opening brace signifies the beginning and the closing brace the end of the compound statement or the block.
- Remember **no semi-colon** should be put after the braces.
- The **null statement** is nothing but simply a semi-colon. There is no statement for any operation.



if...else statement



Syntax

if(condition)

{

//code if condition is true

}

else

{

//code if condition is false

}



if..else...if ladder

if(condition1)

{

//code to be executed if condition1 is true

}

else if(condition2)

{

//code to be executed if condition2 is true

}

else if(condition3)

{

//code to be executed if condition3 is true

}

else{

//code to be executed if all the conditions
are false

}



The if statement vs Conditional operator

- The conditional operator can be used as an alternative to the if statement in some situations.
- The statement **$c = (a < b) ? a : b$** ; is equivalent to the following if statement.

```
if (a < b)
```

```
    c = a;
```

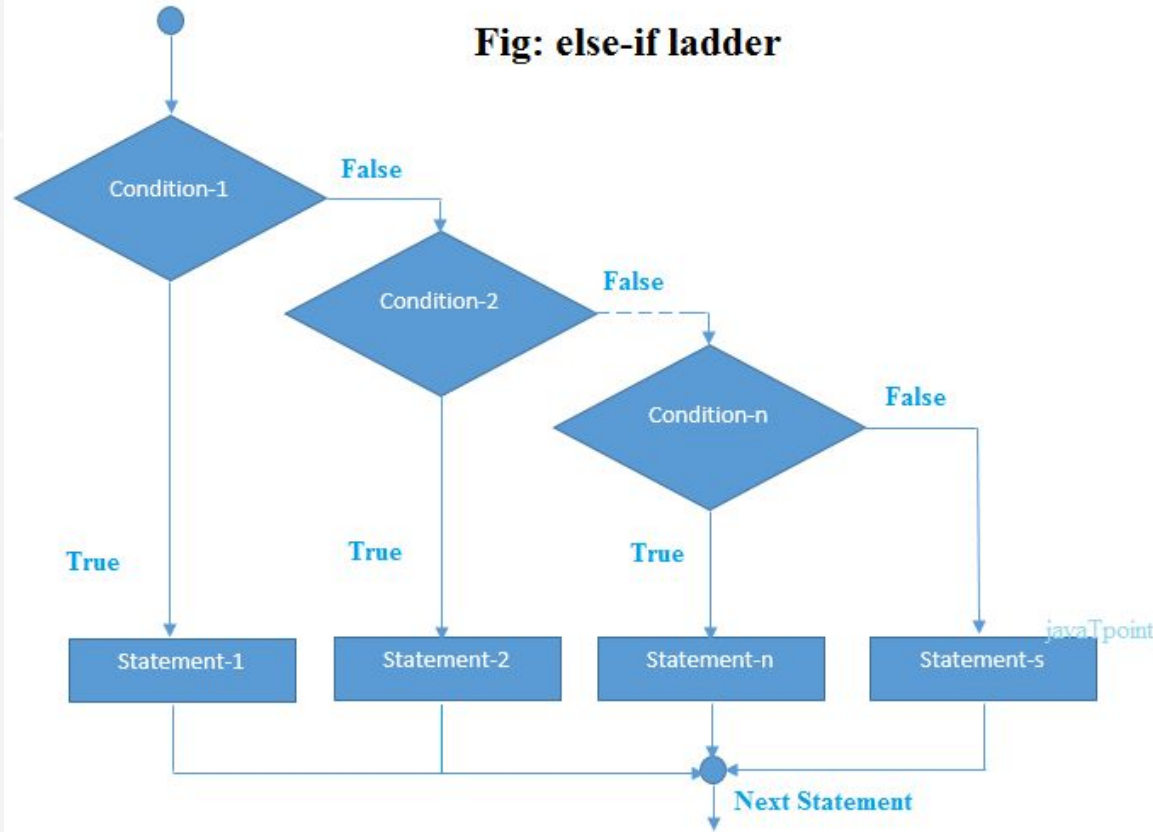
```
else
```

```
    c = b;
```



if..else..if ladder

Fig: else-if ladder



Write a C program to classify the state of an object in SHM based on its position x , amplitude A , and velocity v :

- If $|x| = A$ and $v = 0$: Maximum displacement.
- If $|x| = 0$: At equilibrium.
- If $0 < |x| < A$: In motion.

```
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
    double x, A, v;
    cout << "Enter the amplitude (A): ";
    cin >> A;
    cout << "Enter the position (x): ";
    cin >> x;
    cout << "Enter the velocity (v): ";
    cin >> v;
    if (abs(x) == A && v == 0)
    {
        cout << "The object is at maximum displacement.";
    }
    else if (x == 0)
    {
        cout << "The object is at equilibrium.";
    }
    else if (abs(x) < A)
    {
        cout << "The object is in motion.";
    }
    else
    {
        cout << "Invalid state detected.";
    }
    return 0;
}
```

Write a C++ program to determine the phase of water on its temperature T in Celsius:

- $T < 0$: Solid (Ice).
- $0 \leq T < 100$: Liquid (Water).
- $T \geq 100$: Gas (Steam).

Use **if-else** statements for decision-making.

Write a C++ program to calculate the time of free fall for an object dropped from height h , using the formula:

$$t = \sqrt{\frac{2h}{g}}$$

Where $g = 9.8 \text{ m/s}^2$. Check if:

- The height $h > 0$: Perform the calculation.
- The height $h = 0$: Print "Object is already on the ground."
- The height $h < 0$: Print "Invalid height."


```
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    const double g = 9.8;
    double height, time;

    cout << "Enter the height (in meters): ";
    cin >> height;

    if (height > 0)
    {
        time = sqrt((2 * height) / g);
        cout << "The time for free fall is: " << time << " seconds." << endl;
    } else if (height == 0)
    {
        cout << "The object is already on the ground. Time is zero seconds." << endl;
    } else
    {
        cout << "Invalid height. Height cannot be negative." << endl;
    }

    return 0;
}
```

Write a C++ program to calculate the range of projectile using the formula:
in Celsius:

$$R = \frac{v^2 \sin(2\theta)}{g}$$

Where v is the initial velocity, θ is the angle of projection in degrees, and $g = 9.8 \text{ m/s}^2$. Check:

- If $v \leq 0$: Print "Invalid velocity."
- If $0 \leq \theta \leq 90$: Calculate the range.
- If $\theta > 90$: Print "Invalid angle."

Use a mix of **if-else** and **logical operators** for conditions.

```
#include <iostream>
#include <cmath>

using namespace std;

int main()
{
    const double g = 9.8;
    double velocity, angle, range;

    cout << "Enter the initial velocity (m/s): ";
    cin >> velocity;

    if (velocity <= 0)
    {
        cout << "Invalid velocity. Velocity must be positive." << endl;
        return 1;
    }

    cout << "Enter the angle of projection (degrees): ";
    cin >> angle;
    if (angle < 0 || angle > 90)
    {
        cout << "Invalid angle. Angle must be between 0 and 90 degrees." << endl;
        return 1;
    }

    double angleInRadians = angle * M_PI / 180.0;
    range = (pow(velocity, 2) * sin(2 * angleInRadians)) / g;
    cout << "The range of the projectile is: " << range << " meters.";
```